

ETL Practical Task – Advanced Power BI Data Engineering Simulation

Computer Science Department

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1. Project Setup (Data Engineering Structure)

1. Create a desktop folder: `ETL_Advanced_YourNames_RegNo`
2. Inside the folder, create:
 - 01_Raw_Data
 - 02_Staging
 - 03_Processed
 - 04_Output_Reports
3. Open Power BI Desktop and save as: `Advanced_ETL_Model.pbix`
4. Enable Power Query profiling:
 - Column quality
 - Column distribution
 - Error detection

2. Data Loading (Staging Layer)

5. Load:
 - Sample Store Data
 - Northwind Access Database
6. Load all tables into staging only (do not load to model).
7. Rename staging queries:
 - `stg_orders`
 - `stg_customers`
 - `stg_products`
 - `stg_orderdetails`
 - `stg_users`

3. Data Standardization Layer

Managers Transformation

8. Rename stg_users to dim_managers_stg.
9. Transform columns:
 - Column1 → Region_Code
 - Column2 → Manager_FullName
10. Split FullName into First_Name and Last_Name.
11. Create surrogate key:

$$Manager_Key = HASH(Region_Code + Manager_FullName)$$

12. Clean text: remove spaces, special characters, and standardize casing.

Orders Cleaning

13. Remove: Row ID, Product Base Margin, Order Priority, Quantity Ordered New
14. Filter states: Arizona, California, Florida, New Jersey, New York
15. Rename: State or Province → State
Postal Code → Zip_Code
City → City_Name
16. Sort by State ASC and Sales DESC

4. Data Integration Layer

Orders and Managers Join

17. Perform Left Outer Join on Region_Code.
18. Expand: Manager Name, Region Code
19. Create Manager Performance Class:
 - High (> 10,000)
 - Medium (5,000–10,000)
 - Low (< 5,000)
20. Remove Central region.
21. Sort by Manager Name and Sales.
22. Save as: **fact_orders_enriched**

Northwind Integration

23. Join: Orders → Order Details → Products → Customers
24. Join keys: OrderID, ProductID, CustomerID
25. Create measures:
 - Total Sales = Quantity × Unit Price
 - Discount Adjusted Sales
 - Profit Proxy = Sales × 0.2
26. Remove NULL and orphan records.
27. Save as: **fact_northwind_sales**

5. Business Logic Transformation

28. Convert Discount to percentage (0 decimals).
29. Format Unit Price in USD.
30. Extract from Order Date: Year, Quarter, Month.
31. Remove weekday names.
32. Validate: no duplicates, no NULL keys, correct relationships.

6. Advanced Analytics Preparation

32. Create Customer Segmentation: High, Medium, Low based on Sales.
33. Rank products by sales.
34. Create Regional Performance Index: 50% Sales + 30% Profit + 20% Discount impact.

7. Final Data Warehouse Model

35. Create star schema:
 - fact_orders_enriched
 - fact_northwind_sales
 - dim_managers
 - dim_products
 - dim_customers
36. Ensure: no redundancy, correct relationships, clean keys.

8. Validation Tasks

38. Validate:

- No duplicate primary keys
- No NULL values in fact tables
- Referential integrity maintained
- Correct data types

39. Document all transformation steps.

40. Prepare dataset for Power BI dashboard design.